Momin N. Siddiqui

Georgia Institute of Technology
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Research Statement

My research explores how human-centered AI can support learning, reasoning, and writing. I design intelligent systems that scaffold cognition and augment student thinking to create accessible and reflective learning experiences.

Education

2023 - 2025	**Georgia Institute of Technology**, *Atlanta, GA, USA* M.S. Computer Science (Specialization: HCI), GPA: 4.0/4.0
2019 - 2023	**Jamia Millia Islamia University**, *Delhi, India* B.Tech. Computer Engineering, GPA: 9.41/10

Research Experience

2024 - Present	**Stanford, Institute for Human-Centered AI**, *Research Assistant* Designing and evaluating human-AI interaction methods for collaborative writing tools to support cognitive processes and augment student abilities.
2023 - Present	**Georgia Tech, Teachable AI Lab**, *Graduate Research Assistant* Developing methods and frameworks for Intelligent Tutoring Systems focusing on adaptive scaffolding and agent evaluation.
2022 - 2023	**IIIT Delhi, Human-Machine Interaction Lab**, *Research Intern* Developed computational methods using neuropsychological tests to assess student attention during online learning.

Selected Publications

[1] **Momin N. Siddiqui**, Roy Pea, Hari Subramonyam. "Script&Shift: A Layered Interface Paradigm for Integrating Content Development and Rhetorical Strategy with LLM Writing Assistants." *Proceedings of the CHI Conference on Human Factors in Computing Systems* (CHI 2025). 🏆 **Honorable Mention (Top 5%).**

[2] **Momin N. Siddiqui**, Vryan Feliciano, Roy Pea, Hari Subramonyam. "AI in the Writing Process: How Purposeful AI Support Fosters Student Writing." *Proceedings of the International Conference on Artificial Intelligence in Education* (AIED 2025).

[3] Daniel Weitekamp*, **Momin N. Siddiqui***, Chris MacLellan. "TutorGym: A Testbed for Evaluating AI Agents as Tutors and Students." *Proceedings of the International Conference on Artificial Intelligence in Education* (AIED 2025).

[4] **Momin N. Siddiqui**, Adit Gupta, Jennifer Reddig, Chris MacLellan. "HTN-Based Tutors: A New Intelligent Tutoring Framework Based on Hierarchical Task Networks." *Proceedings of the Eleventh ACM Conference on Learning at Scale* (L@S 2024).

[5] Saumya Yadav, **Momin N. Siddiqui**, Yash Vats, Jainendra Shukla. "EngageME: Exploring Neuropsychological Tests for Assessing Attention in Online Learning." *Proceedings of the International Conference on Artificial Intelligence in Education* (AIED 2024).

(* denotes equal contribution)

Awards

2024	Foley Scholar Award

Invited Talks

2024	**Contextualizing Writing Education for LLM Instigated Paradigm Shift** Georgia Tech IPaT Lunch Lecture, *Guest Lecture*

Leadership and Outreach

2023 – 2024	**Tikkun Olam Makers**, *Project Lead* Led a team of 10 undergraduate students to develop an AAC device for a client with Rhett syndrome.
2019 – 2023	**Tamana Foundation**, *Special Educator* Assisted neurodivergent children with education and rehabilitation needs.

Mentoring

2025	Natalie Hampton, B.S., Stanford CS
2024 – 2025	Vryan Feliciano, M.S., Stanford GSE

Technical Skills

Languages	Python, JavaScript, C++, C#, Bash, Kotlin, HTML/CSS, R, SQL
Libraries	React.js, Node.js, PyTorch, Tensorflow, Docker, Kubernetes, Heroku, Pandas, Scikit-Learn, Numpy, Unity, React Native
Experience	Human-Computer Interaction, NLP, Deep Learning, Knowledge-Based AI, Cognitive Science, Learning Sciences, Multimodal User Interfaces, Web Development, Android Development, Game Development, Backend Programming, User Studies, Data Collection, Annotation, Crowdsourcing